

OASIS INFOBYTE

DATA ANALYTICS INTERNSHIP

LEVEL 2 – TASK 4

GOOGLE PLAY STORE ANALYSIS

Final Project Report

Submitted By

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Internship

OASIS INFOBYTE – Data Analytics Internship

Domain

Data Analytics

Project Title

Level 2 – Task 4: Google Play Store Analysis

Technologies Used

Python • Pandas • NumPy • Matplotlib • Seaborn • TextBlob • Plotly • Jupyter Notebook

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to OASIS INFOBYTE for providing me with the opportunity to complete Level 2 – Task 4, Google Play Store Analysis. This project provided practical exposure to exploratory data analysis, data-quality assessment, duplicate removal, data cleaning, categorical analysis, rating analysis, app-size and install analysis, pricing analysis, revenue estimation, and sentiment analysis.

I also thank the mentors and coordinators for providing the internship task requirements and learning resources. The structured workflow helped me understand how raw application and review data can be cleaned, transformed, visualized, and interpreted to support practical product decisions.

The implementation was completed in a Jupyter Notebook using the Google Play Store Apps dataset and Google Play Store User Reviews dataset. The workflow begins with loading and inspecting both datasets, continues through data cleaning and exploratory analysis, and ends with sentiment analysis by app category and data-driven recommendations for a new app.

This work strengthened my practical understanding of data analytics and demonstrated how category demand, ratings, app size, installs, pricing, estimated revenue, and user sentiment can be combined to derive actionable insights.

ABSTRACT

This project focuses on analyzing Google Play Store application and user-review data to identify patterns that can support the development and positioning of a new mobile application. The executed notebook loads an Apps dataset containing 10,841 records and 13 columns and a User Reviews dataset containing 64,295 records and 5 columns.

The workflow includes dataset inspection, missing-value analysis, duplicate analysis, app-name deduplication, missing-value treatment, conversion of installs and price to numeric values, app-size conversion to megabytes, final quality verification, category distribution analysis, rating analysis, app-size and install analysis, pricing analysis, estimated revenue calculation, and sentiment analysis using TextBlob.

After cleaning, the Apps dataset contains 9,660 records and 14 columns with zero missing values and zero duplicate rows. The review dataset contains 29,692 records after retaining reviews with text, with zero missing values and zero duplicate rows in the final verification. The analysis identifies FAMILY, GAME, and TOOLS as highly represented categories. Most valid ratings are concentrated around the 4.0–4.5 range, while free applications strongly outnumber paid applications.

Sentiment analysis classifies 19,014 reviews as positive, 6,322 as negative, and 4,356 as neutral. Category-level sentiment shows strong positive sentiment in categories such as COMICS and AUTO_AND_VEHICLES, while GAME and SOCIAL exhibit comparatively higher negative sentiment. These results support a product strategy that combines demand analysis with strong user experience and continuous review monitoring.

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CHAPTER 1: INTRODUCTION

1.1 Project Overview

The Google Play Store Analysis project focuses on extracting practical insights from application metadata and user reviews. The project uses application-level attributes such as category, rating, size, installs, type, and price together with review text to understand demand, product quality, monetisation, and user sentiment.

The implementation uses Python, Pandas and NumPy for data handling and transformation, Matplotlib and Seaborn for visualization, TextBlob for sentiment classification, Plotly for interactive category analysis, and Jupyter Notebook for the complete implementation and documentation.

1.2 Problem Statement

A large application marketplace contains many categories, different pricing strategies, varying app sizes, large differences in installation counts, and substantial volumes of user feedback. Raw data also contains missing values, duplicate records, textual numeric fields, and inconsistent representations that must be addressed before analysis.

The project therefore aims to clean the available Google Play Store datasets and identify patterns that can help a developer decide which categories to study, how important ratings are, how users respond to different applications, and how pricing and installs may influence estimated revenue.

1.3 Project Objectives

1. Load and inspect the Google Play Store Apps and User Reviews datasets.
2. Assess dataset structure, missing values, duplicate rows, and duplicate app names.
3. Handle missing values and standardise important numeric fields such as installs, price, and app size.
4. Validate the cleaned datasets and confirm that the final analytical data contains no remaining missing values.
5. Analyze the number of applications across Google Play Store categories.
6. Identify the top 10 most represented categories.
7. Study the distribution of valid app ratings and compare average ratings across categories.
8. Analyze app size, install counts, and the relationship between app size and installs.
9. Compare free and paid applications and study the distribution of paid-app prices.
10. Estimate revenue for paid applications using price multiplied by install count and compare revenue by category.
11. Classify user reviews as positive, negative, or neutral using TextBlob sentiment polarity.
12. Compare sentiment percentages across application categories and identify categories with stronger positive or negative sentiment.
13. Develop data-driven recommendations for a new application based on demand, ratings, monetisation, and user feedback.

CHAPTER 2: DATASET & TOOLS

2.1 Dataset Description

The project uses two Google Play Store datasets. The Apps dataset initially contains 10,841 rows and 13 columns. The User Reviews dataset initially contains 64,295 rows and 5 columns. The Apps data contains application attributes such as App, Category, Rating, Reviews, Size, Installs, Type, Price, Content Rating, Genres, Last Updated, Current Ver, and Android Ver. The review data contains App, Translated_Review, Sentiment, Sentiment_Polarity, and Sentiment_Subjectivity.

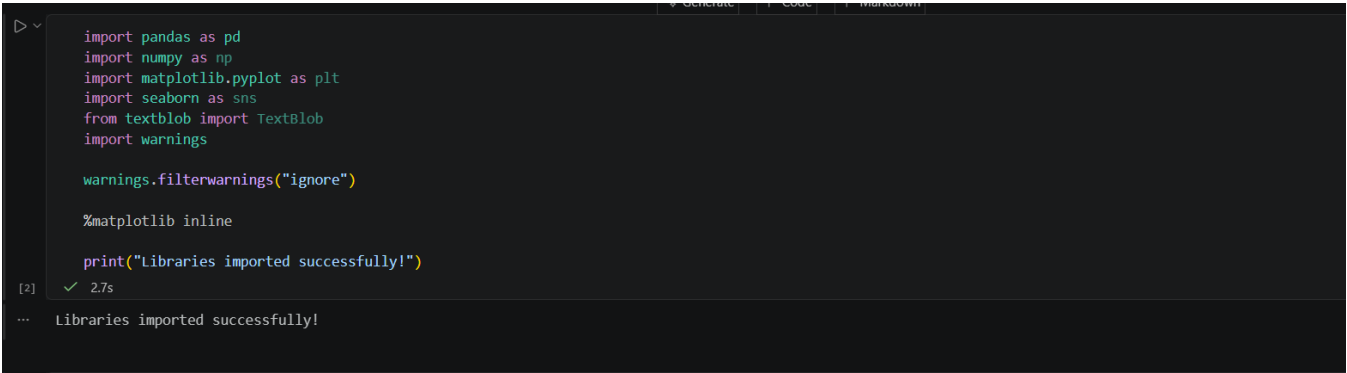
During cleaning, completely duplicated rows were removed from both datasets. The Apps dataset reduced from 10,841 to 10,358 rows after duplicate-row removal and then to 9,660 rows after keeping one record per application name. The Reviews dataset reduced from 64,295 to 30,679 rows after duplicate-row removal and was later reduced to 29,692 review records after retaining non-missing review text.

2.2 Tools & Technologies Used

TOOL / TECHNOLOGY	PURPOSE
Python	Programming language and analytics workflow
Pandas	Dataset loading, cleaning, transformation, grouping, and tabular analysis
NumPy	Numerical operations and supporting calculations
Matplotlib	Charts and analytical visualizations
Seaborn	Statistical and categorical visualizations
TextBlob	Polarity-based user-review sentiment classification
Plotly	Interactive sentiment analysis by app category
Jupyter Notebook	Interactive implementation, execution, and documentation

2.3 Project Folder Structure

The implementation follows a structured project organisation with separate dataset, notebook, screenshots, and report areas. The notebook loads the Apps and User Reviews CSV files from the dataset area and saves analytical figures into the screenshots area. The screenshots folder contains the visual evidence used throughout this report.

A screenshot of a Jupyter Notebook interface. The top part shows a code cell with the following Python code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from textblob import TextBlob
import warnings

warnings.filterwarnings("ignore")

%matplotlib inline

print("Libraries imported successfully!")
```

 Below the code cell, the output is displayed:

```
[2] ✓ 2.7s
... Libraries imported successfully!
```

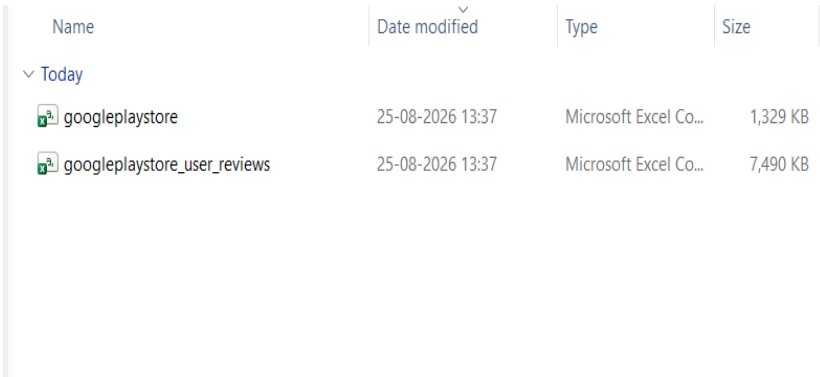
Figure 2.1: Initial Project Execution and Library Import Evidence

Interpretation:

The implementation begins with successful import of the required analytics libraries, establishing the environment for data loading, cleaning, visualization, sentiment analysis, and interactive plotting.

2.4 Input Datasets

The input consists of the Google Play Store Apps dataset and the Google Play Store User Reviews dataset. The initial Apps dataset shape is $10,841 \times 13$, while the initial Reviews dataset shape is $64,295 \times 5$. These datasets provide complementary application metadata and user-feedback information for the analysis.

A screenshot of a file explorer window showing two files under the 'Today' section. The files are 'googleplaystore' and 'googleplaystore_user_reviews', both of which are Microsoft Excel files. The 'googleplaystore' file is 1,329 KB and the 'googleplaystore_user_reviews' file is 7,490 KB. Both files were last modified on 25-08-2026 at 13:37.

Name	Date modified	Type	Size
▼ Today			
 googleplaystore	25-08-2026 13:37	Microsoft Excel Co...	1,329 KB
 googleplaystore_user_reviews	25-08-2026 13:37	Microsoft Excel Co...	7,490 KB

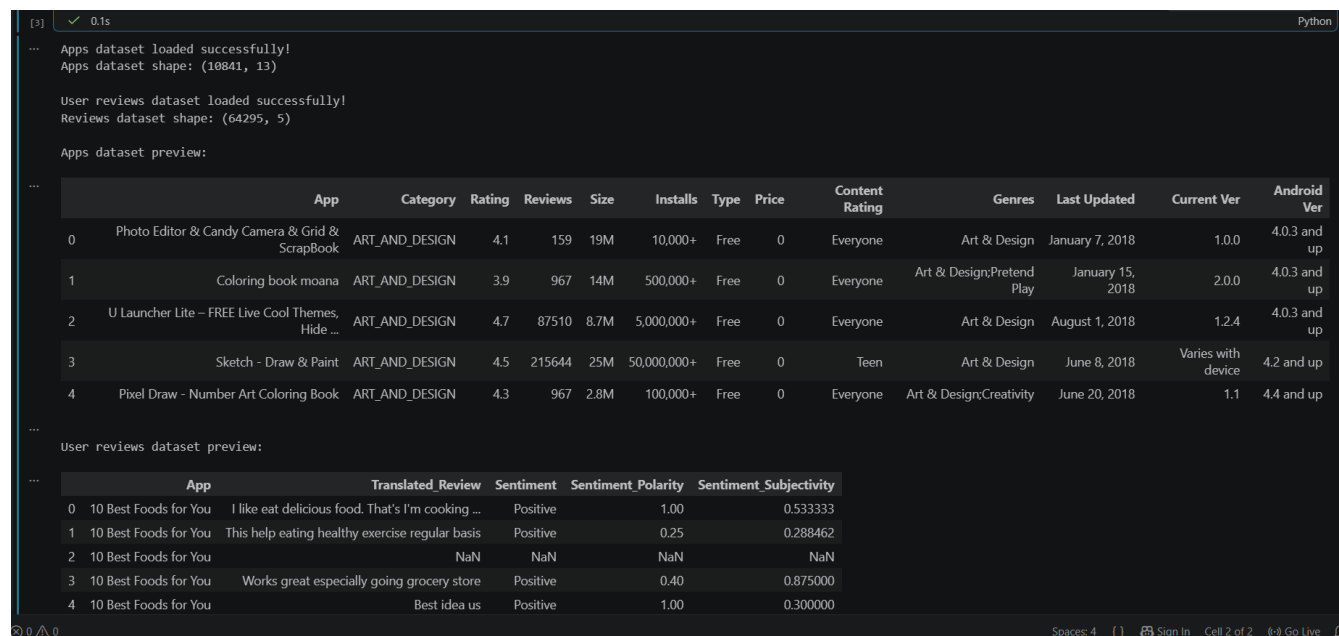
Figure 2.2: Input Datasets - Google Play Store Apps and User Reviews

CHAPTER 3: GOOGLE PLAY STORE ANALYSIS

IMPLEMENTATION

3.1 Library Import & Dataset Loading

The required Python libraries were imported successfully. The Apps dataset and User Reviews dataset were then loaded into Pandas DataFrames using `pd.read_csv()`. The notebook reports 10,841 application records with 13 columns and 64,295 review records with 5 columns.



The screenshot shows a Jupyter Notebook interface with a dark theme. The top bar indicates the execution time is 0.1s. The notebook contains two code cells. The first cell shows the successful loading of the 'Apps' dataset (shape: (10841, 13)) and the 'User reviews' dataset (shape: (64295, 5)). The second cell displays the initial preview of both datasets. The 'Apps' dataset preview shows columns: App, Category, Rating, Reviews, Size, Installs, Type, Price, Content Rating, Genres, Last Updated, Current Ver, and Android Ver. The 'User reviews' dataset preview shows columns: App, Translated_Review, Sentiment, Sentiment_Polarity, and Sentiment_Subjectivity.

Apps dataset loaded successfully!
Apps dataset shape: (10841, 13)

User reviews dataset loaded successfully!
Reviews dataset shape: (64295, 5)

Apps dataset preview:

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Everyone	Art & Design	January 7, 2018	1.0.0	4.0.3 and up
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Everyone	Art & Design;Pretend Play	January 15, 2018	2.0.0	4.0.3 and up
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Everyone	Art & Design	August 1, 2018	1.2.4	4.0.3 and up
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25M	50,000,000+	Free	0	Teen	Art & Design	June 8, 2018	Varies with device	4.2 and up
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2.8M	100,000+	Free	0	Everyone	Art & Design;Creativity	June 20, 2018	1.1	4.4 and up

User reviews dataset preview:

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
0	10 Best Foods for You	I like eat delicious food. That's I'm cooking ...	Positive	1.00	0.533333
1	10 Best Foods for You	This help eating healthy exercise regular basis	Positive	0.25	0.288462
2	10 Best Foods for You	NaN	NaN	NaN	NaN
3	10 Best Foods for You	Works great especially going grocery store	Positive	0.40	0.875000
4	10 Best Foods for You	Best idea us	Positive	1.00	0.300000

Figure 3.1: Dataset Loading and Initial Preview

Interpretation:

Both input datasets were loaded successfully. The Apps dataset provides application-level attributes, while the Reviews dataset provides text and sentiment-related fields for user-feedback analysis.

3.2 Data Inspection & Quality Assessment

The notebook inspected the structure of both datasets and checked missing values. This quality-assessment stage identified fields requiring treatment before downstream analysis and established the initial condition of the raw data.

```
# Check missing values in both datasets

print("==== MISSING VALUES: APPS DATASET =====")
display(apps_df.isnull().sum().sort_values(ascending=False))

print("\n==== MISSING VALUES: USER REVIEWS DATASET =====")
display(reviews_df.isnull().sum().sort_values(ascending=False))

[5] ✓ 0.0s Python
```

```
...
==== MISSING VALUES: APPS DATASET =====
...
Rating          1474
Current Ver       8
Android Ver       3
Content Rating    1
Type              1
Size              0
Reviews           0
Category          0
App               0
Price             0
Installs          0
Last Updated      0
Genres            0
dtype: int64

...
==== MISSING VALUES: USER REVIEWS DATASET =====
...
Translated_Review    26868
Sentiment_Polarity   26863
Sentiment            26863
Sentiment_Subjectivity 26863
App                  0
dtype: int64
```

Figure 3.2: Dataset Quality and Missing-Value Assessment

Interpretation:

The missing-value assessment identifies the columns requiring cleaning and provides the basis for the later imputation and filtering steps.

3.3 Missing-Value Analysis

Missing values were handled in the Apps dataset using appropriate simple treatments. Rating values were filled using the median, while categorical fields such as Current Ver, Android Ver, Content Rating, and Type were filled using their most frequent values. Remaining analytical fields such as Size, Installs, and Price were converted and later completed using median values where required.

```
# Check missing values in both datasets

print("==== MISSING VALUES: APPS DATASET =====")
display(apps_df.isnull().sum().sort_values(ascending=False))

print("\n==== MISSING VALUES: USER REVIEWS DATASET =====")
display(reviews_df.isnull().sum().sort_values(ascending=False))

[5] ✓ 0.0s Python
```

```
... ===== MISSING VALUES: APPS DATASET =====
... Rating          1474
... Current Ver      8
... Android Ver      3
... Content Rating   1
... Type             1
... Size             0
... Reviews          0
... Category         0
... App              0
... Price            0
... Installs         0
... Last Updated     0
... Genres           0
... dtype: int64
...
... ===== MISSING VALUES: USER REVIEWS DATASET =====
... Translated_Review    26868
... Sentiment_Polarity    26863
... Sentiment            26863
... Sentiment_Subjectivity 26863
... App                  0
... dtype: int64
```

Figure 3.3: Missing-Value Analysis Used for Cleaning

Interpretation:

The cleaning workflow converts the raw data into a complete analytical dataset. The final verification later confirms zero missing values in the retained Apps and Reviews data.

3.4 Duplicate Removal

The raw Apps dataset contained 483 completely duplicated rows, while the Reviews dataset contained 33,616 duplicate rows. After removing completely duplicated rows, the Apps dataset contained 10,358 rows and the Reviews dataset contained 30,679 rows. The notebook then checked duplicate application names and found 698 repeated app names.

```
# Remove completely duplicated rows
apps = apps.drop_duplicates().copy()
reviews = reviews.drop_duplicates().copy()

print("\nAfter removing duplicate rows:")
print("Apps dataset shape:", apps.shape)
print("Reviews dataset shape:", reviews.shape)

[6] ✓ 0.0s Python

... Duplicate rows in Apps dataset: 483
Duplicate rows in Reviews dataset: 33616

After removing duplicate rows:
Apps dataset shape: (10358, 13)
Reviews dataset shape: (30679, 5)

Check duplicate app names
rint("Duplicate App names:", apps["App"].duplicated().sum())

Keep one record for each app
pps = apps.drop_duplicates(subset="App", keep="last").copy()

rint("Apps dataset shape after removing duplicate App names:", apps.shape)

[7] ✓ 0.0s Python

... Duplicate App names: 698
Apps dataset shape after removing duplicate App names: (9660, 13)
```

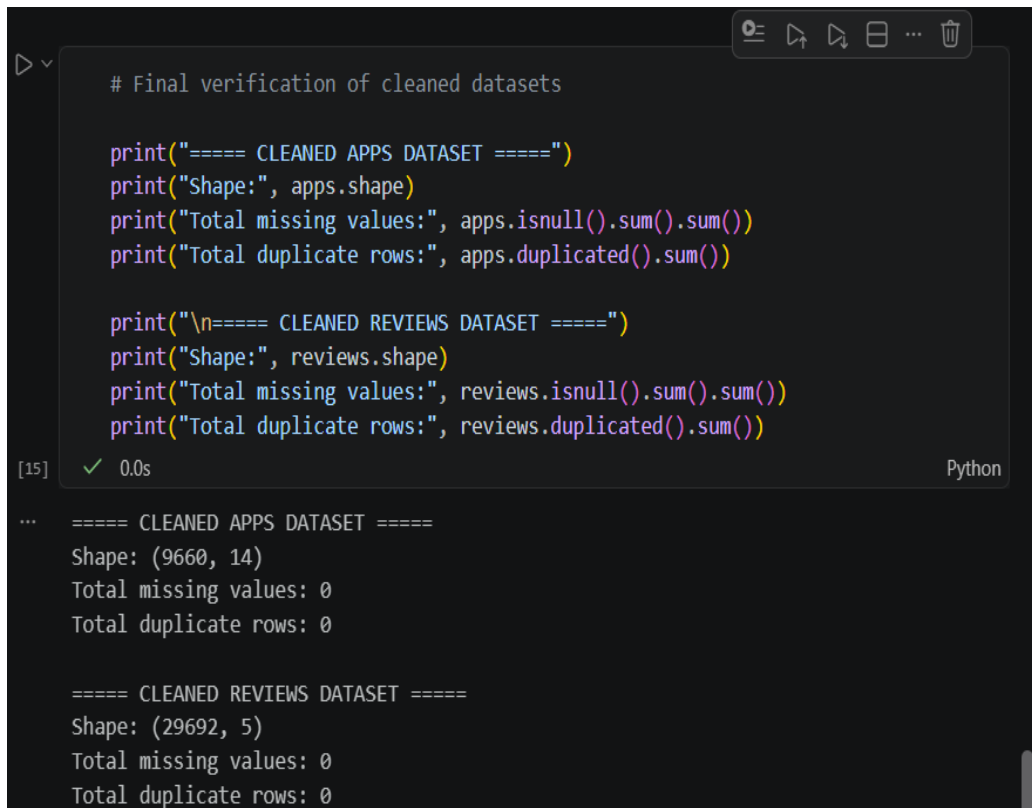
Figure 3.4: Duplicate Detection and Removal

Interpretation:

Duplicate records were explicitly identified and removed. The additional app-name deduplication step ensures that application-level category and rating analyses are not distorted by repeated app records.

3.5 Data Cleaning & Standardisation

The Installs field was cleaned by removing commas and plus signs and converting the result to numeric values. The Price field was converted from currency-formatted text to numeric values. App Size values were converted into megabytes, including conversion of kilobytes to MB. Reviews without review text were excluded before sentiment analysis.



```
# Final verification of cleaned datasets

print("==== CLEANED APPS DATASET ====")
print("Shape:", apps.shape)
print("Total missing values:", apps.isnull().sum().sum())
print("Total duplicate rows:", apps.duplicated().sum())

print("\n==== CLEANED REVIEWS DATASET ====")
print("Shape:", reviews.shape)
print("Total missing values:", reviews.isnull().sum().sum())
print("Total duplicate rows:", reviews.duplicated().sum())
```

[15] ✓ 0.0s Python

```
...
==== CLEANED APPS DATASET ====
Shape: (9660, 14)
Total missing values: 0
Total duplicate rows: 0

==== CLEANED REVIEWS DATASET ====
Shape: (29692, 5)
Total missing values: 0
Total duplicate rows: 0
```

Figure 3.5: Final Data Cleaning Verification

Interpretation:

The final cleaning verification reports an Apps dataset of $9,660 \times 14$ and a Reviews dataset of $29,692 \times 5$, with zero missing values and zero duplicate rows in the final cleaned datasets.

3.6 Category Distribution Analysis

Category frequency analysis shows that FAMILY is the most represented category with 1,906 applications, followed by GAME with 926 and TOOLS with 829. The analysis provides an overview of where application competition is concentrated.

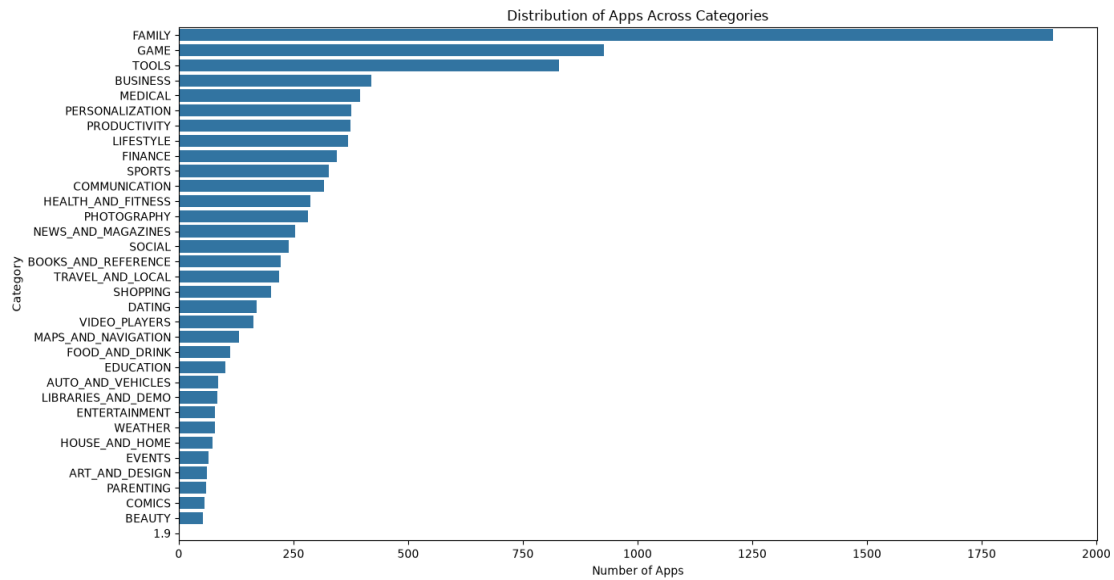


Figure 3.6: Distribution of Apps Across Google Play Store Categories

Interpretation:

The category distribution is highly uneven. FAMILY, GAME, and TOOLS contain substantially more applications than many smaller categories, indicating strong competition and demand concentration.

3.7 Top 10 Saturated Categories

The top 10 categories by application count are FAMILY (1,906), GAME (926), TOOLS (829), BUSINESS (419), MEDICAL (396), PERSONALIZATION (376), PRODUCTIVITY (374), LIFESTYLE (369), FINANCE (345), and SPORTS (327).

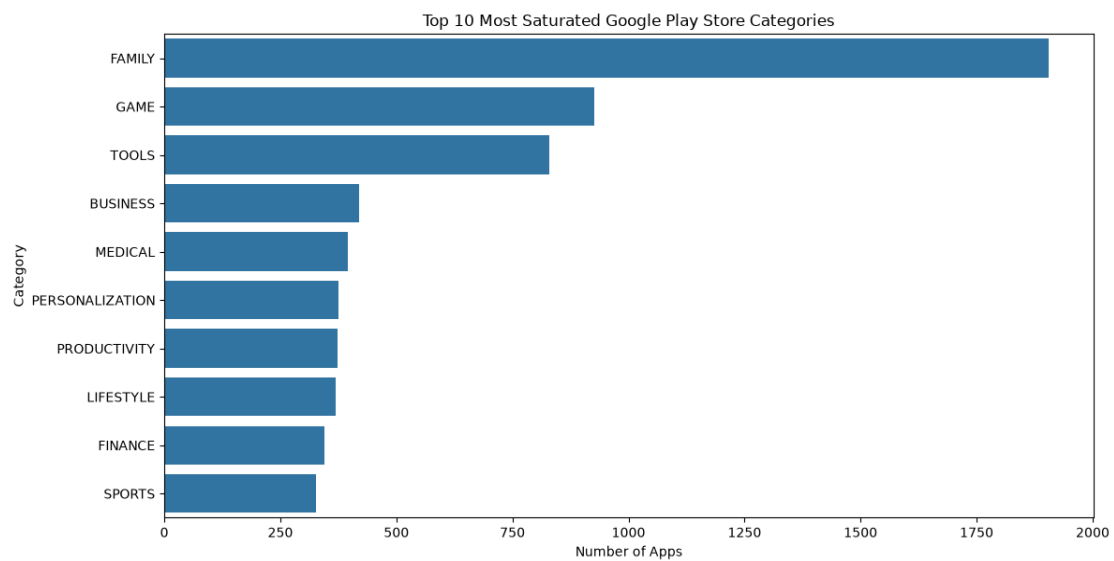


Figure 3.7: Top 10 Most Saturated Google Play Store Categories

Interpretation:

FAMILY is the most saturated category by a wide margin, followed by GAME and TOOLS. A developer entering these categories should identify a clearly differentiated niche rather than competing only on category presence.

3.8 Rating Distribution

For rating analysis, only valid ratings between 1 and 5 were retained. The notebook reports 9,659 valid rating records. The distribution is concentrated toward the higher end of the rating scale, particularly around approximately 4.0–4.5.

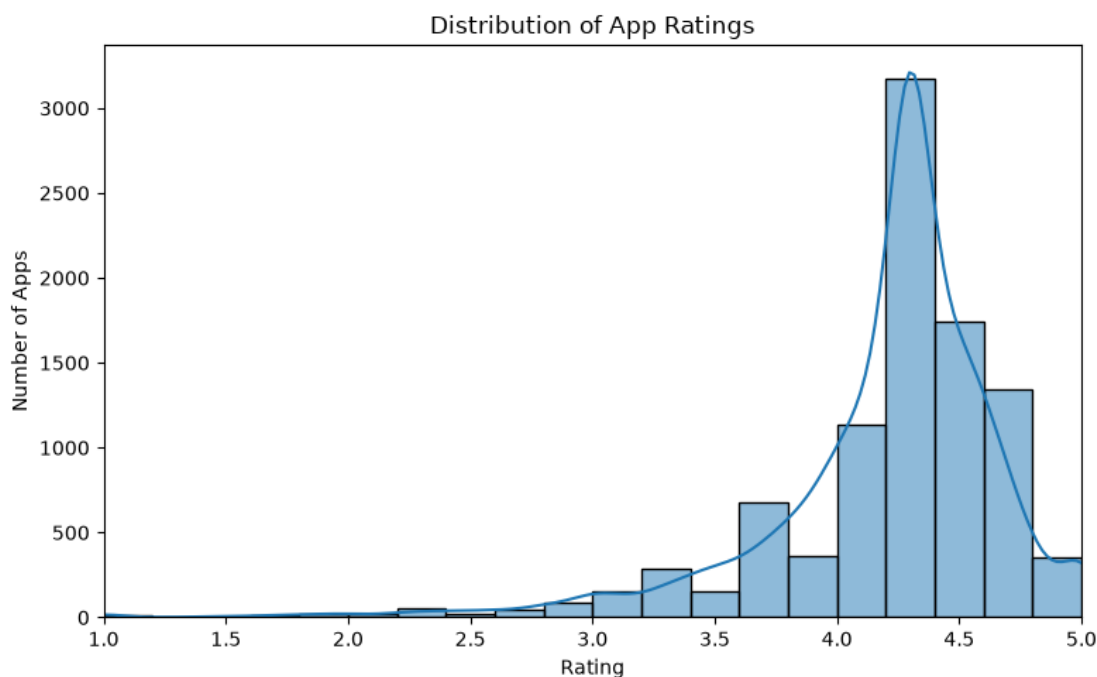


Figure 3.8: Distribution of App Ratings

Interpretation:

Most applications receive ratings in the higher range, showing that strong user satisfaction is common among the retained applications and that maintaining a competitive rating is important.

3.9 Average Rating by Category

Average rating was calculated for each category using valid rating records. The resulting category comparison shows relatively high average ratings across most categories, with differences that can help identify categories where user satisfaction is comparatively strong.

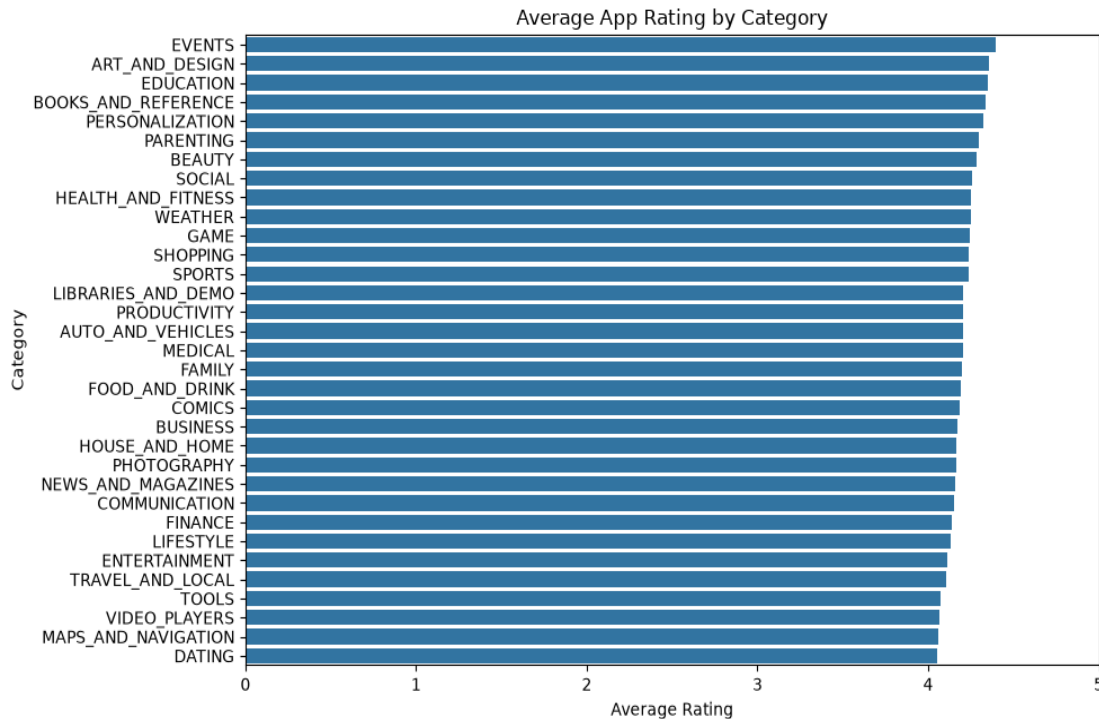


Figure 3.9: Average App Rating by Category

Interpretation:

The category-level averages are clustered around the four-point range. This suggests that a new app should not rely only on entering a popular category; it should also maintain product quality and user satisfaction.

3.10 App Size Distribution

The cleaned Size_MB field contains 8,432 non-missing values. The mean app size is approximately 20.40 MB, with a median of 12 MB and a maximum of 100 MB. The histogram shows that smaller applications are more common, with a long tail toward larger sizes.

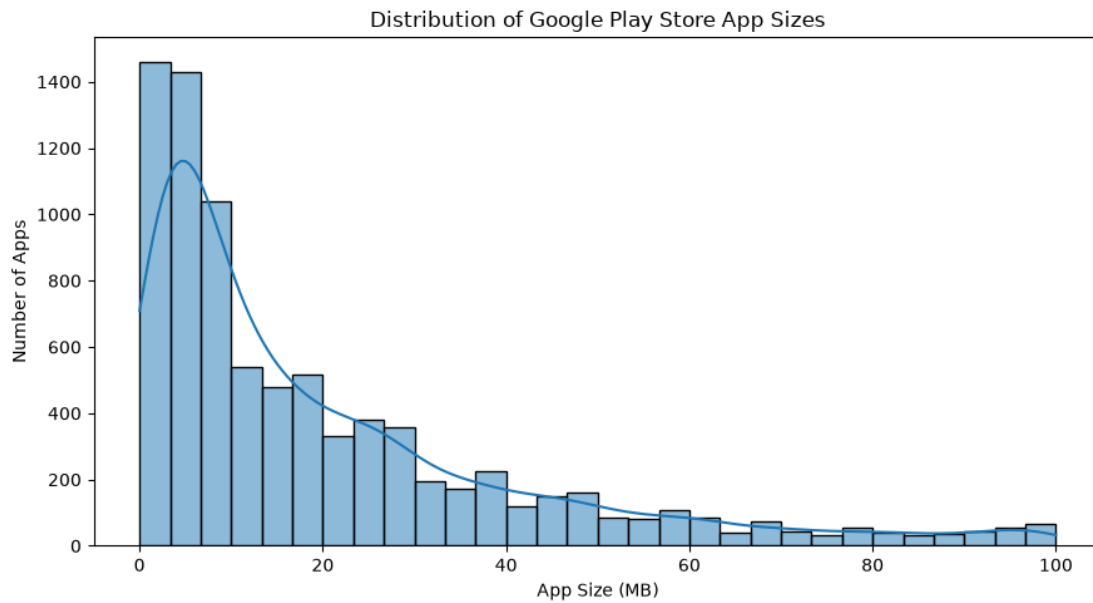


Figure 3.10: Distribution of Google Play Store App Sizes

Interpretation:

The strong concentration of applications at smaller sizes indicates that keeping an application reasonably lightweight may improve accessibility for users with limited storage or network capacity.

3.11 App Installs Distribution

Install counts were converted into numeric values. The cleaned Apps dataset contains 9,660 install observations, with a mean of approximately 7.79 million, a median of 100,000, and a maximum of 1 billion installs. The distribution is highly right-skewed.

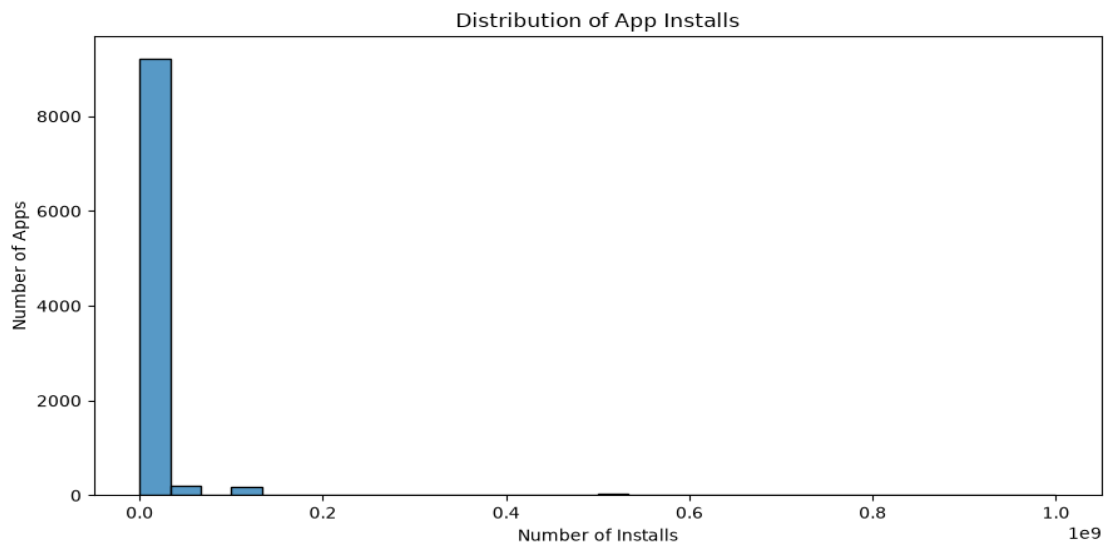


Figure 3.11: Distribution of App Installs

Interpretation:

Most applications have substantially fewer installs than the largest applications. A small number of highly successful apps account for extremely large install counts, making the install distribution highly skewed.

3.12 App Size vs Number of Installs

A scatter plot was used to compare app size in MB with numeric install counts. The visualization shows a wide spread of install levels across different app sizes, including very large install counts for applications of different sizes.

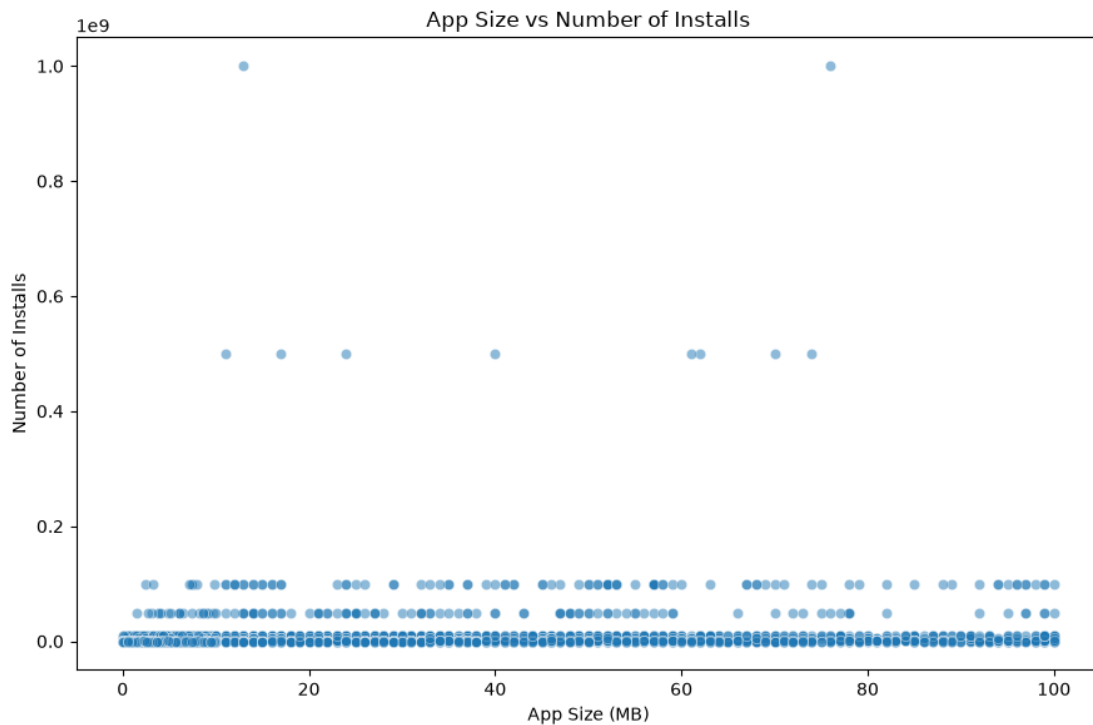


Figure 3.12: App Size vs Number of Installs

Interpretation:

The relationship is not strongly determined by size alone. Applications can achieve high install counts across a range of sizes, indicating that product value, category demand, reputation, and user experience are likely important alongside technical size.

3.13 Pricing Analysis

The numeric Price field was used to classify applications into Free and Paid groups. The final cleaned Apps dataset contains 8,906 free applications and 754 paid applications. Free applications therefore represent the large majority of the dataset.

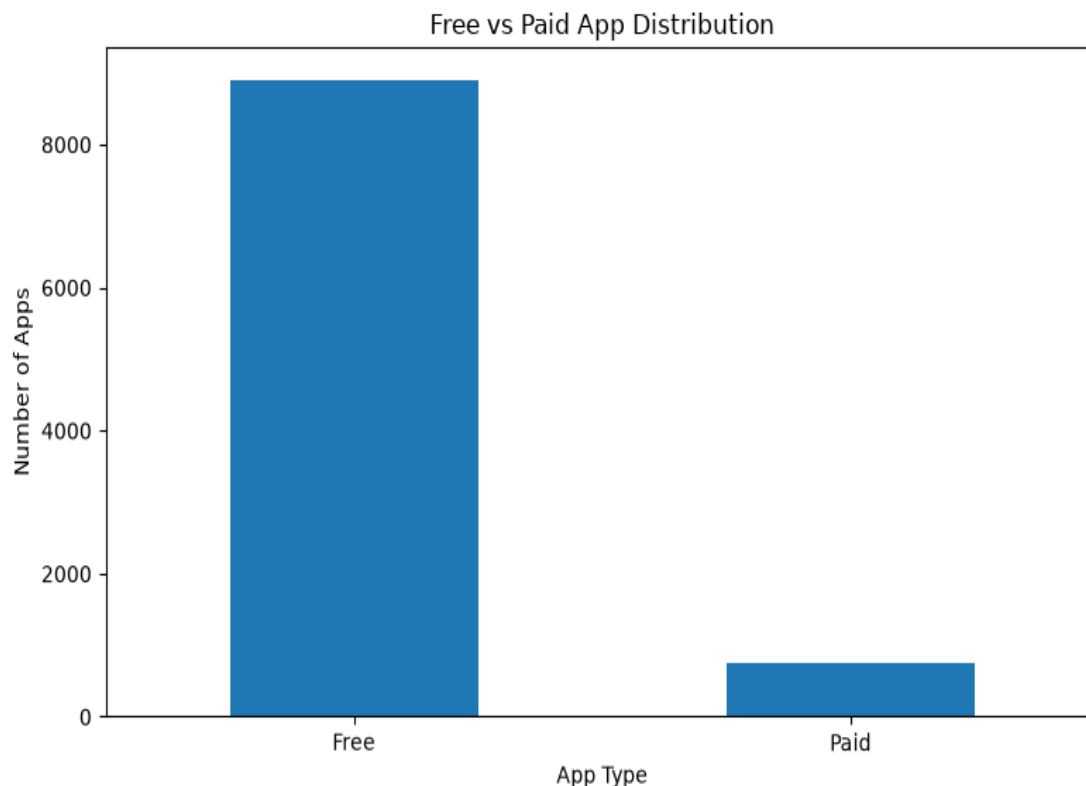


Figure 3.13: Free vs Paid App Distribution

Interpretation:

The marketplace is strongly dominated by free applications. A new developer should therefore treat free access, freemium design, or another low-friction acquisition strategy as an important competitive consideration.

Paid App Price Distribution

The price distribution for paid applications is concentrated at relatively low prices, while a small number of applications have substantially higher prices, including a visible extreme near the upper end of the observed range.

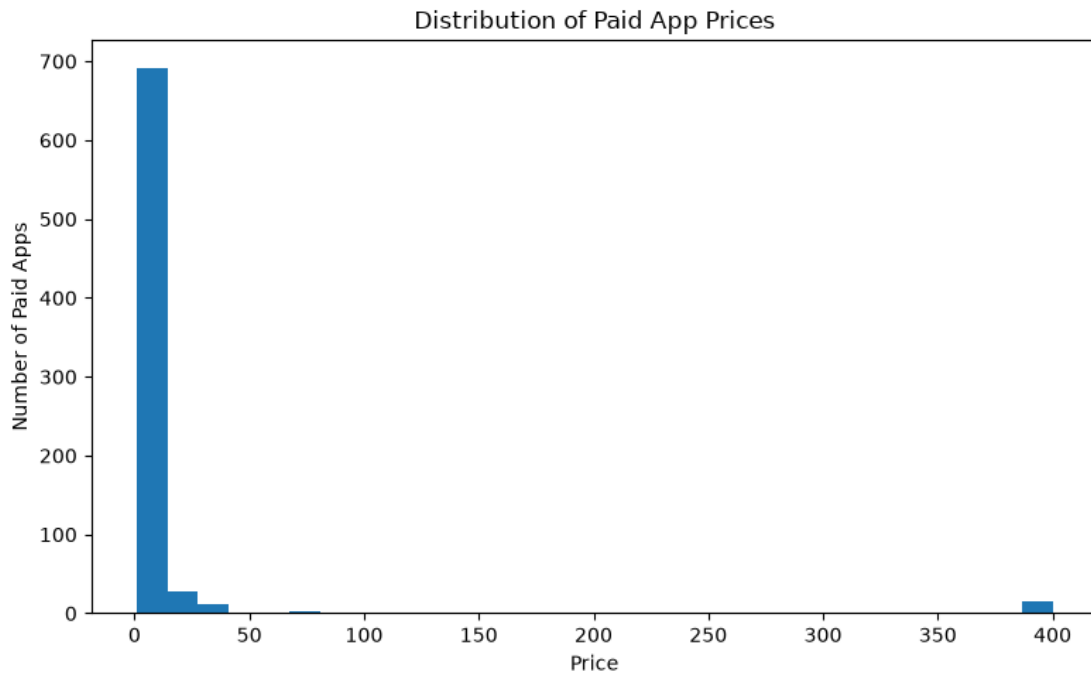


Figure 3.14: Distribution of Paid App Prices

Interpretation:

Most paid applications are priced at lower levels, while a small number of high-priced applications create a long right tail. This indicates that pricing strategy should be aligned with perceived product value and target users.

3.14 Revenue Estimation

Estimated revenue was calculated for paid applications using the implemented formula: Estimated Revenue = Price $\times$ Numeric Installs. The category totals were then grouped and sorted to identify the categories with the highest estimated revenue.

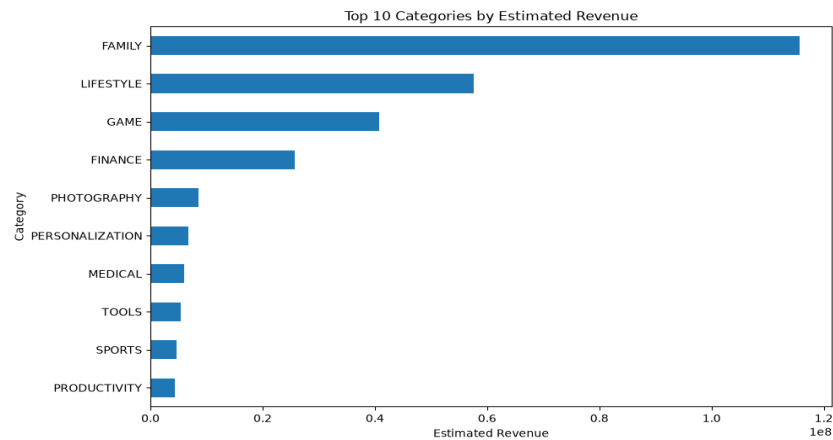


Figure 3.15: Top 10 Categories by Estimated Revenue

Interpretation:

FAMILY has the highest estimated revenue in the executed analysis at approximately 1.16×10^8 , followed by LIFESTYLE, GAME, and FINANCE. The result indicates that application demand and monetisation can combine to create substantially different revenue opportunities across categories.

The revenue estimate is an analytical approximation based on the available price and install figures. It should not be interpreted as actual realised developer revenue because platform fees, taxes, refunds, promotions, and the meaning of install counts are not included in the calculation.

3.15 Sentiment Analysis

The review dataset was prepared by retaining records with non-missing Translated_Review text. TextBlob polarity was then used to classify each review: polarity greater than zero as Positive, less than zero as Negative, and equal to zero as Neutral. The final review dataset contains 29,692 classified reviews.

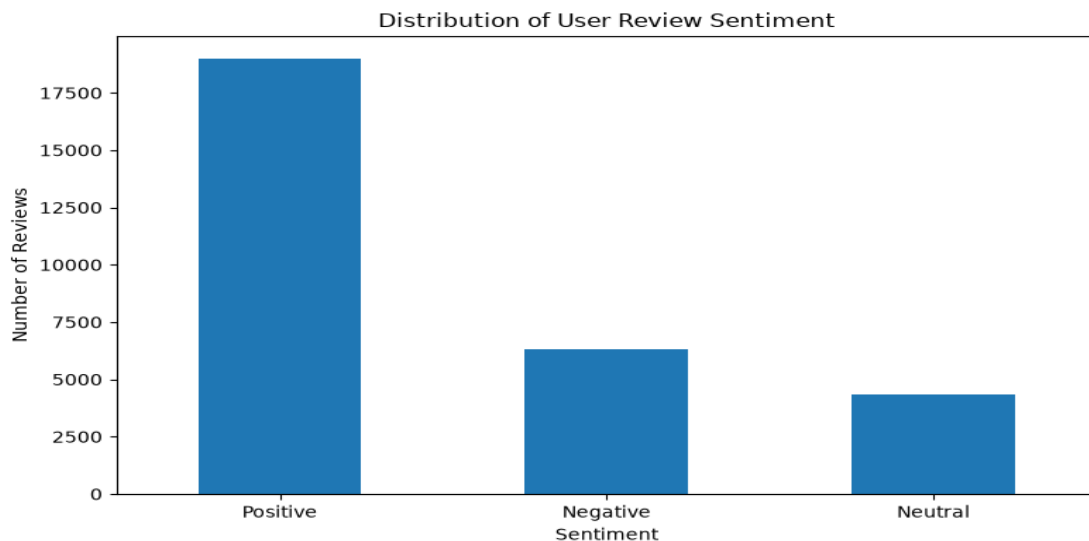


Figure 3.16: Distribution of User Review Sentiment

Interpretation:

The analysis classifies 19,014 reviews as Positive, 6,322 as Negative, and 4,356 as Neutral. Positive feedback is therefore the largest sentiment group, although a substantial number of negative reviews remain available for product-improvement analysis.

The sentiment classification is polarity-based and therefore provides a simple analytical summary rather than a complete semantic understanding of each review. It is most useful for comparing broad sentiment patterns across categories.

3.16 Sentiment Analysis by App Category

The review dataset was merged with application categories using the App field. The merged dataset contains 29,692 records, of which 1,442 reviews have no matching category. For category-level sentiment analysis, reviews without a valid category were excluded.

The category sentiment table calculates the percentage of Positive, Neutral, and Negative reviews within each category and also records the total number of reviews. COMICS has the highest positive sentiment percentage among the displayed categories at 86.67%, followed by AUTO_AND_VEHICLES at 81.27% and EVENTS at 78.85%.

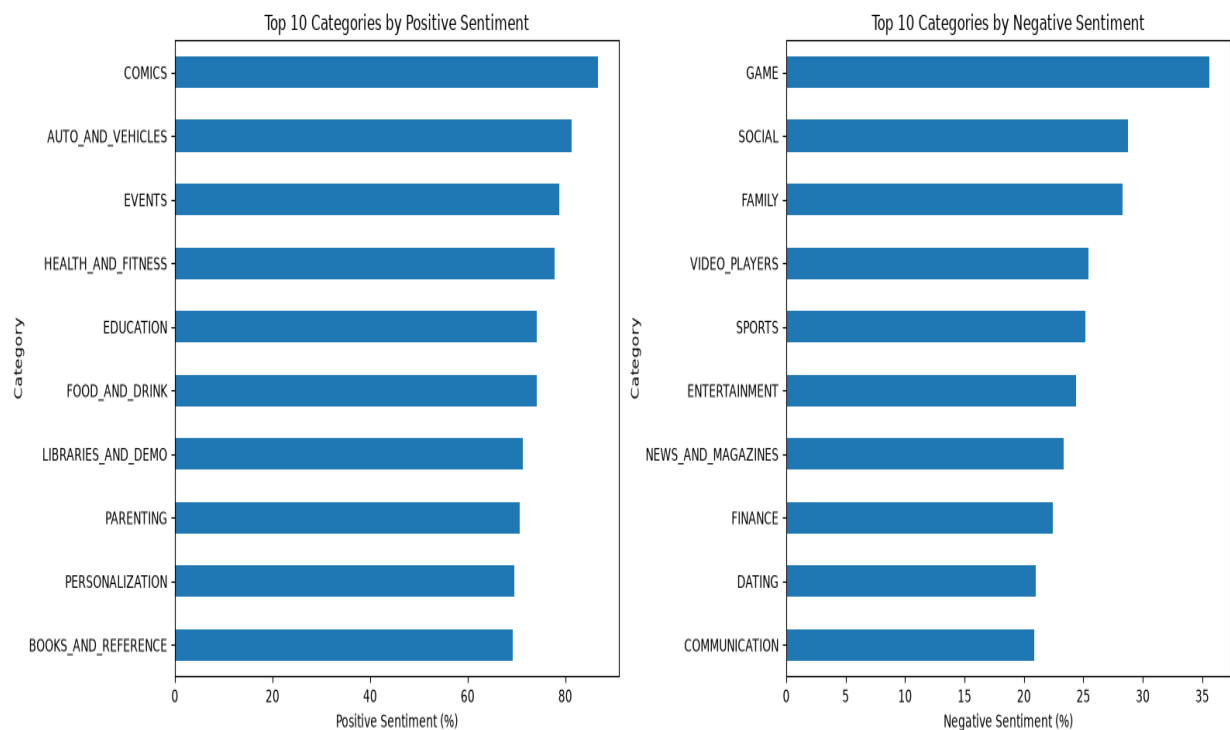


Figure 3.17: Top Categories by Positive and Negative Sentiment

Interpretation:

The comparison highlights different sentiment profiles across categories. COMICS and AUTO_AND_VEHICLES are among the strongest positive categories, while GAME and SOCIAL appear among the categories with comparatively higher negative sentiment.

3.17 Positive and Negative Sentiment Comparison

The top-positive and top-negative category rankings show that user sentiment is not uniform across the marketplace. The strongest positive categories are concentrated around values above approximately 69%, whereas the most negative categories include GAME at about 35.5% negative sentiment and SOCIAL at about 28.8%.

3.18 Interactive Sentiment Analysis

A Plotly-based interactive visualization was created for the top 15 categories by positive sentiment. It compares Positive, Neutral, and Negative sentiment percentages within each selected category, allowing category-level differences to be explored more directly.

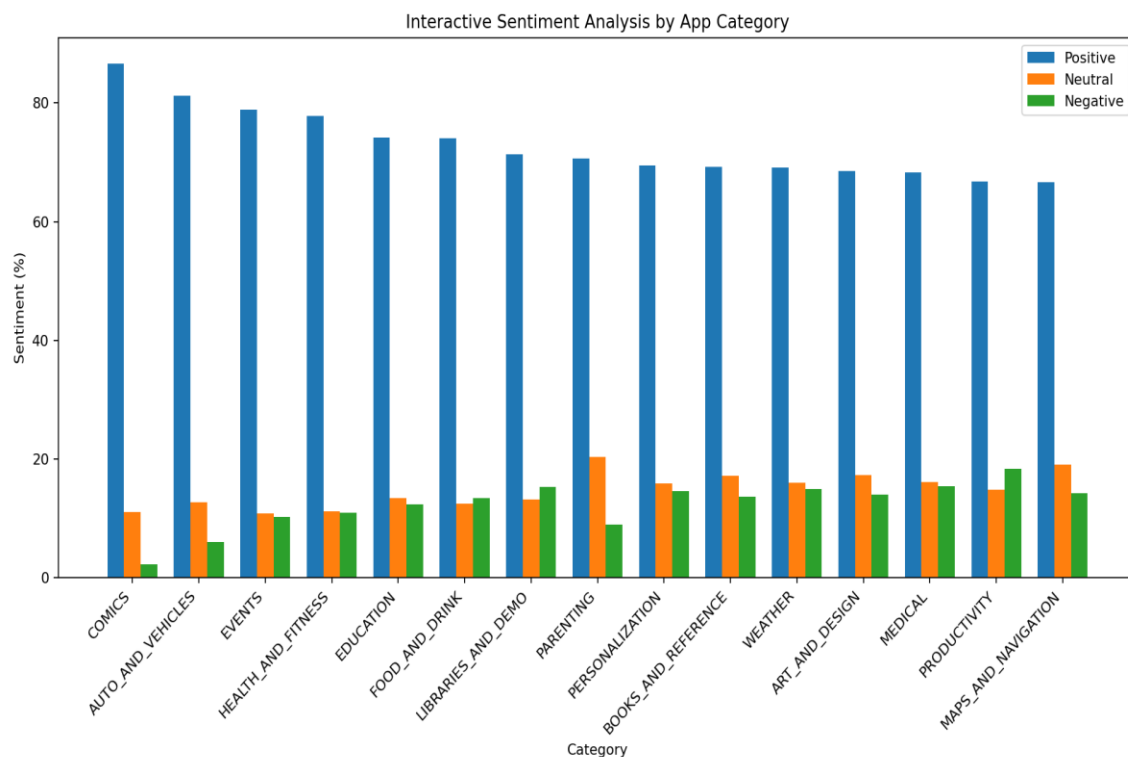


Figure 3.18: Interactive Sentiment Analysis by App Category

Interpretation:

The interactive category view reinforces the main sentiment finding: positive feedback dominates many categories, but the size of the negative and neutral segments varies. Continuous review monitoring can therefore help a developer identify recurring complaints and improve retention.

CHAPTER 4: RESULTS & DISCUSSION

4.1 Key Findings

1. The initial Apps dataset contained 10,841 records and 13 columns, while the initial Reviews dataset contained 64,295 records and 5 columns.
2. 483 duplicate Apps rows and 33,616 duplicate Reviews rows were identified before cleaning.
3. 698 duplicate application names were identified, and one record per app name was retained for the application-level analysis.
4. The final Apps dataset contains 9,660 records and 14 columns with zero missing values and zero duplicate rows.
5. The final Reviews dataset contains 29,692 records with zero missing values and zero duplicate rows after retaining review text.
6. FAMILY, GAME, and TOOLS are the most represented categories, indicating high marketplace concentration.
7. Valid app ratings are strongly concentrated in the upper range, especially around 4.0–4.5.
8. Free applications dominate the dataset with 8,906 apps compared with 754 paid apps.
9. Estimated revenue is highest for FAMILY, followed by LIFESTYLE, GAME, and FINANCE in the executed calculation.
10. TextBlob classifies 19,014 reviews as positive, 6,322 as negative, and 4,356 as neutral.
11. COMICS has the highest displayed positive sentiment percentage, while GAME shows the highest negative sentiment percentage in the top-negative comparison.

4.2 Data-Driven Insights for a New App

1. Focus on high-demand categories: FAMILY, GAME, and TOOLS are among the most represented categories in the Google Play Store. A developer should study these high-demand categories while identifying a specific underserved niche.
2. Prioritize user experience and ratings: Most apps have ratings concentrated around 4.0–4.5, indicating that maintaining a high rating is important for competitiveness. Developers should focus on app quality, reliability, and user satisfaction.
3. Use sentiment to guide product decisions: User reviews are predominantly positive, while categories such as GAME and SOCIAL show comparatively higher negative sentiment. A new app should analyze user feedback continuously and address common complaints to improve retention and reputation.
4. Consider a low-friction acquisition model: Because free applications greatly outnumber paid applications, a new product may benefit from a free or freemium entry model, followed by monetisation through premium features or another value-based strategy.
5. Keep the app reasonably lightweight: The app-size distribution is strongly concentrated at smaller sizes. Efficient application design can reduce storage and download friction, particularly for users with limited device resources.
6. Treat revenue estimates as directional: The estimated revenue analysis is useful for comparing categories but should not be treated as actual financial performance. Platform fees and other commercial factors are outside the executed calculation.

4.3 Conclusion

The Google Play Store Analysis project successfully implemented a complete data-analytics workflow from raw application and review datasets to cleaned data, exploratory visualizations, pricing and revenue analysis, and category-level sentiment insights.

The results show a marketplace with strong concentration in a small number of categories, a predominance of free applications, a rating distribution centred in the higher range, a highly skewed install distribution, and substantial differences in user sentiment across categories. FAMILY is particularly prominent in both application count and estimated revenue in the executed analysis.

The sentiment results show that positive feedback is the dominant class overall, but negative feedback varies meaningfully by category. GAME and SOCIAL are notable areas where comparatively higher negative sentiment suggests that developers should pay close attention to user complaints and product quality.

Overall, the project demonstrates that application-market decisions are stronger when category demand, ratings, technical characteristics, monetisation, and user sentiment are considered together rather than in isolation.

4.4 Future Enhancements

1. Validate category labels and investigate the single anomalous category value observed in the raw category distribution.
2. Use log-scale visualizations for highly skewed install and revenue distributions.
3. Perform statistical correlation and regression analysis between installs, rating, reviews, size, and price.
4. Use more advanced sentiment models capable of handling context, negation, sarcasm, and domain-specific language.
5. Perform topic modelling or keyword extraction on negative reviews to identify recurring product issues.
6. Compare free, freemium, and paid monetisation strategies using additional marketplace or financial data.
7. Use time-aware analysis with Last Updated information to study changes in application activity.
8. Build an interactive dashboard for category, rating, install, pricing, revenue, and sentiment exploration.
9. Validate estimated revenue against real monetisation information where reliable financial data is available.
10. Monitor user sentiment continuously after deployment so product decisions can respond to emerging complaints.

4.5 References

1. Google Play Store Apps dataset and Google Play Store User Reviews dataset used in the submitted Level 2 – Task 4 Jupyter Notebook.
2. Python, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook used for data preparation, analysis, visualization, and documentation.
3. TextBlob used for polarity-based sentiment classification in the submitted implementation.
4. Plotly used for the interactive sentiment analysis visualization.
5. OASIS INFOBYTE Data Analytics Internship – Level 2 Task 4 requirements and implementation workflow.

4.6 Final Findings and Conclusion

The implemented workflow successfully transformed the raw Google Play Store application and review data into a cleaned analytical dataset and produced a set of practical visual and statistical insights. The final Apps dataset contains 9,660 records and 14 columns with no missing values or duplicate rows, while the final review dataset contains 29,692 records with no missing values or duplicate rows.

The analysis indicates that category competition is concentrated in FAMILY, GAME, and TOOLS; user ratings are generally high; free applications dominate the marketplace; app size is strongly concentrated toward smaller values; and install counts are highly skewed toward a relatively small number of highly successful applications.

The revenue estimation shows FAMILY as the leading category in the implemented calculation. The sentiment workflow provides an additional product-quality perspective, with positive reviews dominating overall but category-level negative sentiment varying substantially.

For a new app, the strongest practical strategy supported by this analysis is to identify a differentiated niche within a demand-rich category, maintain a high-quality user experience, keep the application reasonably lightweight, use an accessible monetisation strategy, and continuously analyse user reviews to identify and resolve recurring problems.